THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If a negatively charged rod is brought near a neutral metal disc, the leaves: 1

(a) open **(b)** burn

(c) become charged permanently **(d)** stay closed

2. A small iron nail is attracted by a magnet at 2 cm distance. This shows the force is: 1

(a) friction **(b)** muscular

(c) a contact force **(d)** a non-contact force

3. A box slides faster on a smooth floor than on rough since: 1

(a) friction is smaller on the smooth floor **(b)** air is absent

(c) gravity is smaller **(d)** the mass changes

4. Which sentence is correct? 1

(a) Unlike charges repel **(b)** Charges have no effect

(c) Like charges attract **(d)** Unlike charges attract

5. A floating magnet above another is possible when: 1

(a) no poles **(b)** iron is used

(c) unlike poles face each other **(d)** like poles face each other

6. A force of 6 N and 4 N act on a body in the same direction. The net force is: 1

(a) 1.5 N **(b)** 10 N **(c)** 2 N **(d)** 24 N

7. A car's rubber tyres on a wet road may skid because: 1

(a) water increases friction **(b)** gravity decreases

(c) air is absent **(d)** water reduces friction

8. The speed of a football decreases while it rolls on grass mainly due to: 1

(a) kicking force **(b)** magnetism **(c)** gravity upward **(d)** friction

9. A force of 6 N to the right and 4 N to the left act on a body. The net force is: 1

(a) 2 N to the left **(b)** 2 N to the right **(c)** 10 N **(d)** 0 N

10. Chalk powder on the hands of a gymnast helps to: 1

(a) increase friction **(b)** reduce force

(c) reduce weight **(d)** reduce friction

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are aeroplanes and boats streamlined? 2

12. Why do we put ball bearings in the wheels of a cycle? 2

13. Why do the leaves of an electroscope diverge on touching it with a charged rod? 2

14. Two magnets are placed with like poles facing each other. What do you observe? Why? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A glass rod rubbed with silk is brought near a suspended plastic rod rubbed with wool and they attract. Explain. 3

16. Why is the surface of a ski made smooth and waxed but the tyre of a car made rough? 3

17. Give three examples where we want to reduce friction and three where we want to increase it. 3

OR

Describe how the lightning conductor saves a building. Why is the rod pointed and connected to earth? 3

18. An electroscope is touched by a charged rod. Why does it stay charged but collapse on touching with a hand? 3

OR

How does static electricity affect clothes? Explain the cause and a remedy. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the charging of a comb by rubbing and how it picks up paper. Draw a conclusion about the force involved and why the paper later falls off. 5

OR

List five ways in which we use friction in daily life and five ways of reducing unwanted friction. 5

20. Discuss the roles of gravitational, magnetic and electrostatic force in real life with two examples for each. 5

OR

A man pushes two boxes: one with a smooth bottom and another with a rough bottom, using the same force. Compare their motion and explain the role of friction. 5

END OF QUESTION PAPER — Total: 40 Marks

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